

Arundhati Vasisht

Amherst, MA | 609-937-9405 | avasisht@umass.edu | LinkedIn | GitHub | Portfolio

EDUCATION

University of Massachusetts, Amherst

BS, Computer Science; GPA: 3.83; Chancellor's Award, Dean's List Honoree

Amherst, MA

Expected Fall 2028

SKILLS

Languages: Python, Java, C, TypeScript, JavaScript, SQL, HTML/CSS

Frameworks & Cloud: React.js, Node.js, Express, Flask, TensorFlow/Keras, AWS (Bedrock, Lambda, DynamoDB, OpenSearch), MongoDB, PostgreSQL

AI & Tools: RAG, LLM integration, Qdrant, MediaPipe, Grok, XRPL, Solana/Metaplex, REST APIs, Git

EXPERIENCE

Massachusetts Executive Office of Health & Human Services (EOHHS)

Boston, MA

AI Software Engineer

Spring to Summer 2026

- Built an AI-powered regulatory chatbot for Massachusetts DPH's statewide Metrik platform, targeting 85%+ accuracy across 351 local health departments and 5 inspection domains.
- Architected an AWS Bedrock RAG pipeline grounded in official Massachusetts CMR documents, delivering near-instant regulatory answers while reducing inspectors' 2-10 minute manual document searches.
- Designed statewide audit logging, inspector feedback, and response-flagging workflows to track accuracy, completeness, citations, and retrieval quality across inspector interactions.

Indian Institute of Science (IISc)

Bangalore, India

Machine Learning Research Intern

Summer 2025

- Evaluated 6 ML/DL pipelines across 8,700+ chest X-ray and brain MRI images, implementing SVM, CNN, XGBoost, PCA hybrids, and VGG16 transfer learning for medical image classification.
- Achieved up to 96% test accuracy on brain MRI and 95% on chest X-rays; evaluated accuracy, F1, ROC-AUC, and log-loss across multiple splits while collaborating with 11 researchers.

Minutemen Alternative Investment Fund (MAIF)

Amherst, MA

Quant Arm, Junior Analyst

Fall 2025 to Present

- Built a multi-agent NLP pipeline matching prediction-market contracts across 4 exchanges (Kalshi, Polymarket, Manifold, Metaculus) to identify deterministic arbitrage opportunities.
- Embedded market questions in Qdrant, clustered candidates via KNN, and used Groq/Llama to classify 5 semantic relationships; built live quote polling and alerts for configurable profit thresholds.

PROJECTS

whyrouted | Distributed LLM Load Balancer

Ongoing

- Building a TypeScript load balancer for routing LLM inference traffic across distributed model replicas, implementing 3 scheduling algorithms with 500ms health monitoring to dynamically select healthy, lightly loaded instances.
- Engineering fault-tolerant request handling with automatic replica ejection and 3-attempt failover, achieving <1s failure detection and zero request loss during single-replica failures; validating reliability across 177 automated tests.

KnowYork | Autonomous AI Agent & Blockchain Platform

Fall 2026

- Built a full-stack AI agent that verifies real-world NYC visits and autonomously executes RLUSD rewards across XRPL and Solana, combining Grok vision + multi-sample GPS verification to achieve 90%+ validation accuracy with <4s median decision latency.
- Engineered a zero-trust transaction pipeline with dual-agent verification, deterministic payout controls, and isolated wallet keys, rejecting 95%+ of simulated fraudulent submissions while enforcing a 10 RLUSD on-chain maximum exposure across 25 adversarial security tests.

MedEcho | AI-Powered Medical Results Translator

Spring 2026

- Built a React/Flask system using DenseNet-121 and GradCAM to classify chest X-rays, highlight diagnostic regions, and translate findings into patient-friendly explanations across 4 languages.
- Integrated Gemini, ElevenLabs, and HeyGen LiveAvatar for multilingual result delivery and the ClinicalTrials.gov API for patient-trial eligibility matching; built the complete prototype in 24 hours.

FingerSolve | AI-Powered Hand Gesture Recognition Platform

Spring 2025

- Built a real-time React/Flask gesture-recognition platform for touch-free math quizzes, improving engagement by 40% and reducing user input errors by 30% during pilot testing.
- Engineered MediaPipe Hands + TensorFlow/Keras classification across 12 gestures with 94% accuracy; deployed a prototype that earned 2nd place at the MassAI ML Conference and attracted interest from 50+ educators.

EcoForecast | Full-Stack Sustainability Analytics Platform

Spring 2026

- Built a full-stack React/TypeScript platform simulating 20-year financial ROI and carbon impact of green investments, with Node.js APIs and MongoDB workflows processing quarterly utility data.
- Developed cost, break-even, and carbon-reduction models with 4+ interactive visualizations and integrated a LLaMA 3.3-powered AI advisor for context-aware sustainability recommendations.